

ROMORANTIN-PRUNIER AP, France

WMO: 072470

Lat: 47.3197N

Lon: 1.6875E

Elev: 86

StdP: 100.30

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-4.7	-9.7	1.7	-4.4	-7.1	2.1	-3.0	9.4	11.1	8.5	10.6	0.8	30	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	32.7	20.7	30.4	19.9	28.3	19.3	22.0	30.1	21.0	28.4	20.1	26.6	2.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.4	24.6	18.6	13.5	23.4	17.7	12.8	22.4	64.3	30.1	61.0	28.5	57.7	26.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.9	6.0	-10.4	36.4	3.2	2.6	-12.7	38.3	-14.5	39.8	-16.3	41.2	-18.6	43.1		
			-10.4	23.8	3.2	1.3	-12.7	24.7	-14.6	25.5	-16.4	26.2	-18.8	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	4.3	4.6	7.6	10.3	13.6	17.7	19.2	18.8	15.3	12.0	7.6	4.7
	DBStd	6.45	4.13	3.98	3.55	3.23	3.33	3.51	3.03	3.20	3.06	3.82	3.65	4.13
	HDD10.0	744	179	152	91	35	6	0	0	0	1	24	88	169
	HDD18.3	2694	434	383	333	242	151	53	25	32	99	198	321	423
	CDD10.0	1237	3	2	16	43	118	230	286	273	159	86	17	4
	CDD18.3	145	0	0	0	0	4	33	52	47	8	1	0	0
	CDH23.3	2286	0	0	1	29	114	492	747	702	180	21	0	0
	CDH26.7	827	0	0	0	2	20	180	288	291	44	2	0	0
Wind	WSAvg	2.4	2.7	2.8	2.8	2.6	2.4	2.3	2.3	2.0	1.9	2.1	2.5	2.7
Precipitation	PrecAvg	706	59	51	46	60	67	51	59	52	55	69	70	67
	PrecMax	906	132	88	141	157	174	99	116	106	138	132	140	148
	PrecMin	525	4	6	4	8	21	7	8	8	13	30	23	16
	PrecStd	102	31	25	33	33	31	26	26	30	33	29	30	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	17.6	22.2	26.2	29.1	34.9	35.4	36.8	30.7	25.9	18.5	14.8
		MCWB	11.6	10.7	12.4	15.9	19.8	22.0	20.6	21.8	18.5	17.9	14.5	11.7
	2%	DB	12.8	14.2	19.4	23.6	26.4	31.0	32.3	32.9	27.8	23.0	16.2	13.4
		MCWB	11.0	9.9	11.5	14.7	17.4	20.4	20.9	20.5	18.2	17.0	13.7	11.5
	5%	DB	11.7	12.3	17.0	20.8	24.1	28.5	29.9	30.1	25.4	20.2	14.9	12.3
		MCWB	10.1	9.5	10.9	13.4	16.6	19.8	19.8	19.8	17.5	15.7	12.9	10.7
	10%	DB	10.5	11.0	14.8	18.3	21.8	26.1	27.6	27.3	23.1	18.2	13.5	11.1
		MCWB	9.0	8.9	10.3	12.1	15.3	18.6	19.3	18.9	16.5	14.8	11.5	9.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.0	14.0	17.4	20.6	23.6	22.7	22.8	21.0	18.8	15.5	13.1
		MCDB	13.3	15.0	18.4	24.9	27.1	31.0	30.5	32.1	27.1	23.9	17.0	14.1
	2%	WB	11.4	10.8	12.7	15.4	18.7	21.8	21.7	21.7	19.4	17.5	14.1	11.9
		MCDB	12.5	12.8	16.7	21.8	24.6	28.8	29.9	30.4	25.3	21.3	16.0	13.1
	5%	WB	10.2	10.0	11.7	13.9	17.2	20.5	20.8	20.6	18.3	16.3	12.9	10.8
		MCDB	11.6	12.0	15.3	19.3	22.6	26.8	28.1	28.0	23.9	19.7	14.7	12.1
	10%	WB	9.1	9.0	10.9	12.6	15.9	19.2	19.7	19.6	17.2	15.2	11.8	9.8
		MCDB	10.4	10.9	14.4	17.3	20.9	24.9	26.3	25.7	21.8	17.7	13.3	11.0
Mean Daily Temperature Range	5% DB	MDBR	7.0	9.1	11.9	13.0	12.8	13.0	13.6	14.5	14.2	11.2	8.2	7.2
		MCDBR	6.9	12.0	17.3	18.7	17.2	17.8	18.7	20.1	18.5	14.0	9.7	8.3
		MCWBR	5.4	8.3	10.2	10.3	9.2	8.4	8.2	8.6	9.6	8.6	7.1	6.6
	5% WB	MCDBR	6.3	9.5	12.9	16.4	15.2	15.5	16.8	17.1	14.9	12.3	9.1	7.2
		MCWBR	5.3	7.0	8.2	9.5	8.6	7.8	7.7	8.2	8.3	7.8	6.9	6.1
Clear-Sky Solar Irradiance	taub	0.323	0.343	0.372	0.401	0.404	0.416	0.410	0.399	0.368	0.364	0.336	0.320	
	taud	2.445	2.389	2.314	2.232	2.274	2.276	2.290	2.336	2.404	2.422	2.447	2.429	
	Ebn at Noon	758	817	844	850	861	850	849	844	840	781	733	712	
	Edh at Noon	68	87	109	130	129	130	126	116	99	84	67	62	
All-Sky Solar Radiation	RadAvg	1.04	1.88	3.10	4.44	5.22	5.89	5.71	5.02	3.89	2.34	1.26	0.90	
	RadStd	0.11	0.26	0.38	0.54	0.48	0.57	0.37	0.46	0.28	0.20	0.14	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page